

ON THE GEOMETRY OF NONLINEAR ORDINARY DIFFERENTIAL EQUATIONS

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CHAPTER 1

THE WRONSKIAN, TOPOLOGY AND HARMONIC MOTION

In this section we shall quickly develop the foundations that we shall base our future analysis upon. This includes some basic linear algebra of $\mathcal{C}([a, b])$ spaces and simple Wronskian theory together with metric space topology. At the end of this chapter, we shall apply what we have done to solve a problem in mechanics. In the above, we denote by

$$\mathcal{C}([a, b]) := \{f : [a, b] \rightarrow \mathbb{C} : f \text{ is continuous on } [a, b]\} \quad (1.1)$$

where the interval $[a, b] \subset \mathbb{R}$ is given.

1.1 THE SPACE OF CONTINUOUS FUNCTIONS AND THE WRONSKIAN MATRIX

Recall the following definition:

DEFINITION 1. *Given a set X and a function $d : X \times X \rightarrow [0, \infty)$ we call the pair (X, d) a metric space if*

1. $d(x, x) = 0$ for all $x \in X$.
2. $d(x, y) = d(y, x)$ for all $(x, y) \in X \times X$.
3. $d(x, y) \leq d(x, z) + d(z, y)$ for all $(x, y, z) \in X^3$.

In this case, we call d a metric on X .

A metric space is called complete if each Cauchy sequence has a limit point in X . That is, a metric space is complete when a sequence converges if and only if it is Cauchy. Continuity is defined in the usual way, under the metric topology induced by the metric d . A function $\Phi : X \rightarrow X$ is called a *contraction* provided there exists a constant $0 < C < 1$ so that

$$d(\Phi(x), \Phi(y)) \leq Cd(x, y), \quad \forall (x, y) \in X \times X$$

Given any function $f : X \rightarrow X$ we call a point $\xi \in X$ a fixed point of f if $f(\xi) = \xi$. A crucial and elementary result in the analysis of metric spaces is the following fixed point theorem:

THEOREM 1.1.1 (Banach Fixed Point Theorem). *Let (X, d) be a complete metric space and $\Phi : X \rightarrow X$ a contraction on the space. There exists a unique fixed point $\xi \in X$.*

Proof. We first show uniqueness. Suppose by way of contradiction that $f(\xi) = \xi$ and $f(\zeta) = \zeta$ but $\xi \neq \zeta$. Then, we have $d(\xi, \zeta) > 0$ as well. Now,

$$0 < d(\xi, \zeta) = d(f(\xi), f(\zeta)) \leq Cd(\xi, \zeta)$$

implying that $C \geq 1$ which is absurd.

Having established this, we now show existence. Fix a generating point $\xi_0 \in X$. Define the following recursive sequence:

$$\xi_n := \begin{cases} \Phi(\xi_{n-1}), & n \geq 1 \\ \xi_0, & n = 1 \end{cases} \quad (1.2)$$

this is clearly well defined. Now we claim this sequence is Cauchy. Certainly, by the triangle inequality we have for $n, m \geq N$ with $n > m$:

$$d(\xi_n, \xi_m) \leq d(\xi_n, \xi_{n-1}) + d(\xi_{n-1}, \xi_{n-2}) + \cdots + d(\xi_{m+1}, \xi_m)$$

Where we see for any positive integer j : $d(\xi_j, \xi_{j-1}) \leq C^{j-1}d(\xi_1, \xi_0)$. Therefore, we have the following estimate on $d(\xi_n, \xi_m)$:

$$\begin{aligned} d(\xi_n, \xi_m) &\leq (C^{n-1} + C^{n-2} + \cdots + C^m) d(\xi_1, \xi_0) = C^m d(\xi_1, \xi_0) \sum_{j=0}^{n-m-1} C^j \\ &= C^m d(\xi_1, \xi_0) \frac{1 - C^{n-m}}{1 - C} \\ &\leq \frac{C^N}{C - 1} d(\xi_1, \xi_0) \end{aligned}$$

Thus, for any $\varepsilon > 0$ one may take N so that $\frac{C^N}{C - 1} d(\xi_1, \xi_0) < \varepsilon$. Then, if $n, m \geq N$ it follows from the above estimate that $d(\xi_n, \xi_m) < \varepsilon$. Hence (ξ_n) is Cauchy in a complete metric space and must have a limit point $\xi \in X$. We shall now show that this is a fixed point of Φ . Indeed, a contraction is uniformly continuous by definition and therefore continuous. So, it follows that

$$\Phi(\xi_n) \xrightarrow{n \rightarrow \infty} \Phi(\xi)$$

However, $\xi_{n+1} = f(\xi_n)$ for all n and as $n \rightarrow \infty$ we must have as well $\xi_n \rightarrow \xi$, which is the same sequence as ξ_{n+1} . Therefore,

$$\xi = \lim_{n \rightarrow \infty} \xi_{n+1} = \lim_{n \rightarrow \infty} \Phi(\xi_n) = \Phi(\xi)$$

which completes the proof. ■

Now we shall rapidly develop the theory of Wronskian matrices. Recall that given a vector space $(V, \oplus, \odot)$ over a field $\mathbb{F}$ we say a finite family of vectors $\{u_j\}_j$ are linearly independent if

$$a_1 u_1 \oplus a_2 u_2 \oplus \cdots \oplus a_n u_n, \quad a_j \in \mathbb{F}$$

implies that $a_j = 0_{\mathbb{F}}$ for all j . In this same vector space, we say a subset $\mathcal{B}$ of V is a basis for V if the elements in $\mathcal{B}$ are linearly independent and each vector $u \in V$ may be expressed as a linear combination of the elements in $\mathcal{B}$. Here, we shall develop a useful test for the linear independence of infinite dimensional vectors.

DEFINITION 2. Let $F := (f_n)_{n=1}^N \subset \mathcal{C}([a, b])$ be a finite collection of vectors and $[a, b] \subset \mathbb{R}$. The Wronskian associated to F , denoted $\mathcal{W}[F]$, is a function

$$\mathcal{W}[F] : [a, b] \rightarrow \mathbb{C}^{N \times N}$$

determined by

$$\mathcal{W}[F](x) := \begin{pmatrix} f_1(x) & f_2(x) & \cdots & f_N(x) \\ f_1'(x) & f_2'(x) & \cdots & f_N'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(N-1)}(x) & f_2^{(N-1)}(x) & \cdots & f_N^{(N-1)}(x) \end{pmatrix} \quad (1.3)$$

when these derivatives exist.

The surprising result is that

THEOREM 1.1.2. *Let $F := (f_n)_{n=1}^N \subset \mathcal{C}([a, b])$ be a finite collection of vectors and suppose the Wronskian matrix $\mathcal{W}[F]$ exists on $[a, b] \subset \mathbb{R}$. If there exists a point $x_0 \in [a, b]$ such that $\det(\mathcal{W}[F](x_0)) \neq 0$ then F is a linearly independent family.*

Proof. Suppose not, then we may find constants $a_1, a_2, \dots, a_N$ in $\mathbb{C}$ with at least one non-zero and

$$a_1 f_1(x) + a_2 f_2(x) + \cdots + a_N f_N(x) = 0$$

successive differentiation yields the following system of equations:

$$a_1 f_1(x) + a_2 f_2(x) + \cdots + a_N f_N(x) = 0 \quad (1.4)$$

$$a_1 f_1'(x) + a_2 f_2'(x) + \cdots + a_N f_N'(x) = 0 \quad (1.5)$$

$$\vdots \quad (1.6)$$

$$a_1 f_1^{(N-1)}(x) + a_2 f_2^{(N-1)}(x) + \cdots + a_N f_N^{(N-1)}(x) = 0 \quad (1.7)$$

This holds, in particular at $x = x_0$. Now we write this in matrix form to recover:

$$\mathcal{W}[f](x_0) \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Since $\mathcal{W}[F](x_0)$ is invertible, we conclude that $a_1, a_2, \dots, a_N = 0$ which is a contradiction. This completes the proof. ■

LEMMA 1.1.3 (Superposition Principle). *Let f_1, f_2 be solutions to $L[y] = 0$ for a linear differential operator L . Then any superposition of f_1 and f_2 also solves $L[y] = 0$.*

Proof. Let $a, b \in \mathbb{C}$ be fixed. Simply write,

$$L[af_1 + bf_2] = aL[f_1] + bL[f_2] = 0$$
■

1.2 SIMPLE HARMONIC MOTION

In this section we consider a simple system equipped with a spring. Suppose that we have a mass m attached to a spring and we shall choose a frame of reference so that the spring is attached to the origin. Suppose the mass experiences acceleration a (e.g gravity), pulling it and stretching the spring.

NONLINEAR ANALYSIS OF ODEs

Let $x(t)$ denote the displacement from the “resting position” (the spring is completely coiled) at a time $t \geq 0$ and denote $x(0) = x_0$. Denote the spring constant by $k > 0$. Recall *Hooke’s Law* which then states that

$$F = -kx \quad (1.8)$$

where F is the force acting upon the spring at the position x . By Newton’s second law of motion we have $F = ma$ and thence

$$ma + kx(t) = 0 \quad (1.9)$$

or, since $k \neq 0$:

$$a + \omega^2 x(t) = 0, \quad \omega := \sqrt{\frac{k}{m}} > 0$$

now, note that $a = x''(t)$ whence we recover the differential equation:

$$x''(t) + \omega^2 x(t) = 0, \quad t \geq 0 \quad (1.10)$$

This leads us to the study of the following initial value problem. Suppose we start the system with the spring at a distance of $x_0 \geq 0$, i.e. we assume $x(0) = x_0$, what is the behaviour of the system over time? To answer this, we seek to solve the following IVP:

$$\begin{cases} x''(t) + \omega^2 x(t) = 0, & t \geq 0 \\ x(0) = x_0 \\ x'(0) = u_0 \end{cases} \quad (1.11)$$

Given particular constants one may be able to numerically analyze the behaviour of the system, but this is certainly not the perfect solution. It is natural to wish for a “well behaved” solution in closed form. Since we do not wish to limit ourselves to particular constants, we shall construct a basis $\mathcal{B}$ for the space of all solutions, where we let these variables k, m, a vary.

The strategy is as follows. Suppose we take the function $x(t) = e^{\alpha t}$ for some constant α yet to be determined. Certainly, this function is entire and there is no trouble “plugging” it into the differential equation to try and determine a suitable constant α . Doing this yields,

$$\alpha^2 e^{\alpha t} + \omega^2 e^{\alpha t} = 0$$

or, equivalently

$$\alpha^2 + \omega^2 = 0$$

since $e^x \neq 0$ for all x . The two possible choices of α are then $\alpha = \pm \omega i$. It is left to the reader to verify that indeed

$$f_1(t) = e^{\omega i t}, \quad f_2(t) = e^{-\omega i t}$$

solve the differential equation. Now, to see that these indeed form a basis for the space of solutions note:

$$\det(\mathcal{W}[f_1, f_2](t)) = \det \begin{pmatrix} e^{\omega i t} & e^{-\omega i t} \\ \omega i e^{\omega i t} & -\omega i e^{-\omega i t} \end{pmatrix} = -2\omega i \neq 0$$

This is good, we have solved the differential equation, as we can always find suitable constants $a, b \in \mathbb{C}$ so that

$$f(t) = ae^{\omega i t} + be^{-\omega i t}$$

satisfies (1.11). However, we began with real variables. It would be nice to finish with a solution from the reals into the reals. By Euler's identity, we may write

$$\begin{aligned}f_1(t) &= \cos \omega t + i \sin \omega t \\f_2(t) &= \cos \omega t - i \sin \omega t\end{aligned}$$

Now, by the *Superposition Principle* we see that $g_1(t) = \frac{1}{2}f_1(t) + \frac{1}{2}f_2(t) = \cos \omega t$ is also a solution to the differential equation. The same argument applies to $\frac{1}{2i}f_1(t) - \frac{1}{2i}f_2(t) = g_2(t) = \sin \omega t$. We claim that these are still independent. Certainly,

$$\det(\mathcal{W}[g_1, g_2](t)) = \det \begin{pmatrix} \sin \omega t & \cos \omega t \\ \omega \cos \omega t & -\omega \sin \omega t \end{pmatrix} = -\omega \neq 0$$

Therefore $\{g_1, g_2\}$ forms a basis for the space of solutions. Then, for appropriate constants $A, B \in \mathbb{C}$ we solve (1.11) via the superposition

$$A \sin \omega t + B \cos \omega t = \gamma \cos(\omega t + \delta)$$

by the sum of sin and cos formula. Here γ represents the amplitude of the generated wave, and δ can always be solved for.

CHAPTER 2

A TOPOLOGICAL APPROACH TO EXISTENCE AND UNIQUENESS

We devote this section to the proof of the Picard-Lindelöf Theorem, which establishes the existence of a continuously differentiable solution to a particular, general class, of first order differential equations. We recall first that a function is *Lipschitz continuous* if $|f(x) - f(y)| \leq L|x - y|$ for a positive constant L and appropriate x, y . In particular, Lipschitz continuity implies uniform continuity.

This is not too great of a restriction, functions that are continuously differentiable on a compact interval are Lipschitz by the Mean Value theorem. The theorem we wish to prove is the following:

THEOREM 2.0.1 (Picard-Lindelöf). *Denote by Ω a neighbourhood of (u_0, x_0) such that*

$$\Omega := \{(x, u) \in \mathbb{R}^2 : |x - x_0| < a, |u - u_0| < b\} \quad (2.1)$$

Let $\Gamma : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function that is Lipschitz continuous in the second variable throughout Ω . Then, the initial value problem

$$\begin{cases} u'(x) = \Gamma(x, u) \\ u(x_0) = u_0 \end{cases} \quad (2.2)$$

There exists $d > 0$ so that the problem above has a unique solution in $[x_0 - h, x_0 + h]$.

As we shall see, the proof is quite elegant, but far from trivial. This is a common theme with this sort of problem. Existence is a very difficult topic, and we shall be seeing one of the easiest proofs of this kind. It is a beautiful, intricate proof but we still require many preliminary facts which we give in the following subsection.

2.1 LEMATTA

DEFINITION 3. *Let $(V, \oplus, \odot)$ be a vector space over $\mathbb{C}$ and $\|\cdot\| : V \rightarrow [0, \infty)$ a norm on the space $(V, \oplus, \odot)$. The 4-tuple $(V, \oplus, \odot, \|\cdot\|)$ is called a **normed vector space** over $\mathbb{C}$. If d is the metric:*

$$d_V : V \times V \rightarrow [0, \infty), \quad (u, v) \mapsto \|u - v\|$$

*induced by the norm, and the metric space (V, d_V) is complete, the 4-tuple $(V, \oplus, \odot, \|\cdot\|)$ is called a **Banach space**.*

Let $[a, b] \subset \mathbb{R}$ be a compact interval. We define again

$$\mathcal{C}([a, b]) := \{f : [a, b] \rightarrow \mathbb{C} : f \text{ is continuous on } [a, b]\}$$

We shall now endow this space with a norm. Consider the following norm:

$$\|\cdot\|_\infty := \sup_{x \in [a, b]} |\cdot(x)| < \infty \quad (2.3)$$

It is left to the reader to convince themselves that this turns $\mathcal{C}([a, b])$ into a normed vector space over $\mathbb{C}$ when we equip the space with the usual addition and scalar multiplication. The following is going to be crucial in our proof:

PROPOSITION 2.1.1. *$(\mathcal{C}([a, b]), +, \cdot, \|\cdot\|_\infty)$ is a Banach space.*

Proof. Let $(f_n) \subseteq \mathcal{C}([a, b])$ be a Cauchy sequence. Then, for each $\varepsilon > 0$ there exists $N \in \mathbb{N}$ so that whenever $n, m \geq N$ one has $\|f_n - f_m\|_\infty < \varepsilon$. In particular, for each $x \in [a, b]$ one sees that $|f_n(x) - f_m(x)| < \varepsilon$ whenever $n, m \geq N$. Thus, the sequence $(f_n(x))$ is Cauchy in $\mathbb{C}$ for each x .

Since $\mathbb{C}$ is complete, there exists a complex number $f(x)$ so that $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$, for any x . We show that $f(x) \in \mathcal{C}([a, b])$ and $f_n \rightarrow f$ with respect to the norm $\|\cdot\|_\infty$. We begin with convergence in the norm. Let $\varepsilon > 0$, since the sequence is Cauchy we may find N so large that

$$\sup_{x \in [a, b]} |f_n(x) - f_m(x)| < \varepsilon, \quad \forall n, m \geq N$$

whence $|f_n(x) - f_m(x)| < \varepsilon$ for all x . Letting $m \rightarrow \infty$ we recover $|f_n(x) - f(x)| \leq \varepsilon$ for all $n \geq N$ and $x \in [a, b]$. Therefore,

$$\|f_n - f\|_\infty = \sup_{x \in [a, b]} |f_n(x) - f(x)| \leq \varepsilon, \quad \forall n \geq N$$

which shows that f_n converges to f in this space. Now, to see that $f \in \mathcal{C}([a, b])$ let $\varepsilon > 0$ and fix $x \in [a, b]$. Since the sequence is again Cauchy we may find N_1 so that

$$|f_n(x) - f_m(x)| < \varepsilon, \quad \forall x \in [a, b], \quad n, m \geq N$$

Now, since $\lim_{m \rightarrow \infty} f_m(x) = f(x)$ we may find $N_2 > 0$ so that $|f_n(x) - f(x)| < \varepsilon$ whenever $n \geq N_2$. Set $N := \max(N_1, N_2)$ and let $n, m \geq N$ to deduce

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| \\ &< 2\varepsilon + |f_n(x) - f_n(y)| \end{aligned}$$

Now, since f_n is continuous at x for any $x \in [a, b]$ and $n \in \mathbb{N}$ we may find $\delta > 0$ so that $|f_n(x) - f_n(y)| < \varepsilon$ whenever $|x - y| < \delta$. For this $\delta > 0$, in the above we find

$$|f(x) - f(y)| < 3\varepsilon$$

which completes the proof. ■

LEMMA 2.1.2. *Let (X, d) be a complete metric space and $Y \subseteq X$ a closed set. Denote by d_Y the restriction of d to Y . Then, (Y, d_Y) is a complete metric space.*

Proof. If (y_n) is Cauchy in Y then it is Cauchy in X , which is complete and hence there exists a point $y \in X$ so that $\lim y_n = y$. But since Y is closed and each $y_n \in Y$ it follows immediately that $y \in Y$. ■

COROLLARY 2.1.3. *Given $b > 0$ set*

$$\mathcal{C}_b([a, b]) := \{f \in \mathcal{C}([a, b]) : \|f - x_0\|_\infty \leq M\}$$

This is a complete metric space under the ∞ -norm.

Proof. Since $\mathcal{C}_b([a, b]) \subset \mathcal{C}([a, b])$ it is sufficient to show that $\mathcal{C}([a, b])$ is closed. Without loss of generality we take $x_0 = 0$. Let (f_n) be a sequence of elements in $\mathcal{C}_b([a, b])$ converging to f with respect to the norm $\|\cdot\|_\infty$. Suppose that $f \notin \mathcal{C}_b([a, b])$, so there exists $x \in [a, b]$ so that $|f(x)| > b$.

Now, since $f_n \rightarrow f$, by choosing $\varepsilon := |f(x)| - b > 0$ we may find $N \in \mathbb{N}$ so that $\|f_N - f\|_\infty < |f(x)| - b$. That is,

$$|f_N(x) - f(x)| < |f(x)| - b$$

whence

$$-|f_N(x)| < -b \iff b < |f_N(x)|$$

which implies $\|f_N\|_\infty > b$, a contradiction. ■

We now wish to give a complete characterization to C^1 solutions of (2.2). The first thing to note is that this differential equation and initial value problem may be reformulated in terms of an integral equation. Indeed, suppose a function u solves (2.2) on some interval $[x_0 - h, x_0 + h]$, then:

$$u(x) = u(x_0) + \int_{x_0}^x \Gamma(t, u) dt \tag{2.4}$$

completely describes this solution u . Conversely, we note that any function of the form in (2.4) solves the initial value problem (2.2).

Having made these observations we note that it is sufficient to show the existence of a unique solution to the integral equation in (2.4). We shall invoke the so-called “fixed point method”, which sometimes works on higher order ODEs and even PDEs, although modern methods for existence with regards to PDEs are much more complex and involve Sobolev spaces, which are beyond the scope of this text.

The astute reader should observe that the derived form in (2.4) can be used whenever we only have one present differential, certainly consider the IVP:

$$\begin{cases} u^{(n)} \equiv \Gamma(x, u) \\ u^{(n-1)}(x_0) = u_{n-1}, \dots, u(x_0) = u_0 \end{cases}$$

In this case $n = 2$, this becomes

$$\begin{cases} u'' \equiv \Gamma(x, u) \\ u'(x_0) = v_0 \\ u(x_0) = u_0 \end{cases} \tag{2.5}$$

Then, suppose u is a C^2 solution of the IVP above. We may write for appropriate x , bounded by a translation of h ,

$$u'(\sigma) = v_0 + \int_{x_0}^{\sigma} \Gamma(t, u) dt$$

hence,

$$u(x) = u_0 + v_0(x - x_0) + \int_{x_0}^x \int_{x_0}^\sigma \Gamma(t, u) dt d\sigma \quad (2.6)$$

completely characterizes solutions to (2.5).

The reader should be able to derive a characterization for all C^n solutions based on the argument used above for the general case, which we do not write explicitly here. In practice, the cases $n = 1, 2$ are by far the most useful and frequently encountered.

2.2 PROOF OF THE PICARD-LINDELÖF THEOREM

Proof of The Picard-Lindelöf Theorem. Define the operator $I[\cdot]$ given by

$$I[\cdot] := u_0 + \int_{x_0}^x \Gamma(t, \cdot) dt$$

acting upon $\mathcal{C}_b([x_0 - h, x_0 + h])$ for all $h > 0$ and let Ω be as given in the statement of the theorem. To ease notation, we shall write $V_h(x_0)$ to denote $[x_0 - h, x_0 + h]$. We shall refine this mapping so that

$$\mathcal{C}_b(V_h(x_0)) \ni u \mapsto I(u) \in \mathcal{C}_b(V_h(x_0))$$

Now, fix x, y and let $u \in \mathcal{C}_b(V_h(x_0))$, without loss of generality taking $y < x$

$$\begin{aligned} |I[u](x) - I[u](y)| &= \left| \int_{x_0}^x \Gamma(t, u) dt - \int_{x_0}^y \Gamma(t, u) dt \right| = \left| \int_y^x \Gamma(t, u) dt \right| \\ &\leq \int_y^x |\Gamma(t, u)| dt \\ &\leq M |x - y| \end{aligned}$$

where $M := \sup_\Omega |\Gamma|$, thereby showing that for any $u \in \mathcal{C}_b(V_h(x_0))$ the function $I[u]$ is uniformly continuous, and hence is $\mathcal{C}(V_h(x_0))$. We now shift our attention to the IVP. In order to apply Theorem (1.1.1) we shall make I a contraction on $\mathcal{C}_b(V_h(x_0))$ for $h > 0$ yet to be determined. Let h be given, note for a pair $(u, v) \in \mathcal{C}_b(V_h(x_0))^2$ one has over Ω :

$$\begin{aligned} |I[u](x) - I[v](x)| &= \left| \int_{x_0}^x \Gamma(t, u) dt - \int_{x_0}^x \Gamma(t, v) dt \right| \leq \int_{x_0}^x |\Gamma(t, u) - \Gamma(t, v)| dt \\ &\leq L \int_{x_0}^x |u(t) - v(t)| dt \\ &\leq L \int_{x_0}^x \|u - v\|_\infty dt \\ &\leq Lh \|u - v\|_\infty \end{aligned}$$

since x was arbitrary it follows that $\|I[u] - I[v]\|_\infty \leq Lh \|u - v\|_\infty$. To ensure that this is a contraction, we shall require that $Lh < 1$ or, $h < \frac{1}{L}$. Now, we observe that

$$|I[u](x) - u_0| = \left| \int_{x_0}^x \Gamma(u, t) dt \right| \leq \int_{x_0}^x |\Gamma(u, t)| dt \leq Mh \leq b$$

where we require this last inequality, so that we map back into the required subregion $\mathcal{C}_b(V_h(x_0))$. That is, we additionally require $h \leq \frac{b}{M}$. Now, taking

$$h := \min \left\{ \frac{1}{L}, \frac{b}{M} \right\}$$

The mapping I is a contraction on $\mathcal{C}_b(V_h(x_0))$, and we know from Corollary (2.1.3) that $\mathcal{C}_b(V_h(x_0))$ is complete with respect to the ∞ -norm. Therefore, by Theorem (1.1.1) we conclude there exists a unique fixed point to (2.4), and hence the IVP.

This concludes the proof. ■

COROLLARY 2.2.1. *Under the same conditions, there exists a unique solution to (2.5) for h sufficiently small.*

CHAPTER 3

ANALYSIS OF A QUASILINEAR ODE

In this section we examine the one dimensional *Nonlinear Stationary Schrödinger Equation*. In $\mathbb{R}^d$, this corresponds to the following problem. Let $\Omega \subseteq \mathbb{R}^d$ be a convex domain, we seek to solve the following boundary value problem:

$$\begin{cases} -\Delta u + u \equiv u^3 & \text{in } \Omega \\ u \equiv 0 & \text{on } \partial\Omega \\ u > 0 & \text{in } \Omega \end{cases} \quad (3.1)$$

where Δ denote the Laplacian:

$$\Delta := \sum_{\ell=1}^d \frac{\partial^2}{\partial x_\ell^2}$$

We shall study the one-dimensional problem:

$$\begin{cases} u'' = g(u) & \text{in } \Omega \\ u \equiv 0 & \text{on } \partial\Omega \\ u > 0 & \text{in } \Omega \end{cases}$$

where $g : \mathbb{R} \rightarrow \mathbb{R}$ locally Lipschitz. That is, given any point $x \in \mathbb{R}$ there exists a neighbourhood $\mathcal{U}$ of x so that g is Lipschitz with constant $L > 0$ in $\mathcal{U}$. Note that this a generalization of the original statement. Before we proceed to the main instances of the problem, we give a nice covering lemma:

LEMMA 3.0.1. *Let $g : \mathbb{R} \rightarrow \mathbb{R}$. Then g is locally Lipschitz on $\mathbb{R}$ if and only if it is Lipschitz for each compact interval $I \subset \mathbb{R}$.*

Proof. ($\Leftarrow$) If g is Lipschitz for each compact interval $[a, b] \subset \mathbb{R}$ then given $x_0 \in \mathbb{R}$ take the interval $(x_0 - 1, x_0 + 1)$. Since the closure of this interval is compact, g is Lipschitz there, and hence also on $(x_0 - 1, x_0 + 1)$.

($\Rightarrow$) Conversely, we argue by contradiction. Suppose instead that $g \in \text{Lip}_{\text{Loc}}(\mathbb{R})$ but $g \notin \text{Lip}([a, b])$. Then, for each $L > 0$, there exists a pair $(x, y) \in I \times I$ so that $|g(x) - g(y)| > L|x - y|$; and hence $x \neq y$ for any such pair. By setting $L = n \in \mathbb{N}$ we construct associated sequences $(x_n), (y_n)$ in I with

$$n < \frac{|g(x_n) - g(y_n)|}{|x_n - y_n|}$$

Note that we may extract a subsequence $(x_{n_k}) \rightarrow x \in I$. For all such indices we preserve:

$$n_k < \frac{|g(x_{n_k}) - g(y_{n_k})|}{|x_{n_k} - y_{n_k}|}$$

Now, g is continuous on $\mathbb{R}$ and hence it is bounded on I . So, as $n_k \rightarrow \infty$ we require $|x_{n_k} - y_{n_k}| \rightarrow 0$, as is easily seen. This precisely means that $(y_{n_k}) \rightarrow x$ as well. Now, there is a small enough neighbourhood $\mathcal{U} \ni x$ and a positive K so that

$$|g(z) - g(w)| \leq K |z - w|, \quad \forall z, w \in \mathcal{U}$$

So,

$$\frac{|g(z) - g(w)|}{|z - w|} \leq K, \quad \forall z, w \in \mathcal{U}$$

By taking n_k very large we can ensure that $x_{n_k}, y_{n_k} \in \mathcal{U}$, but then this would contradict the above. ■

Before we proceed with particular cases, there are several remarks to be made about the underlying differential equation

$$u'' + g(u) \equiv 0, \quad g \in \text{Lip}_{loc}(\mathbb{R}) \quad (3.2)$$

where $u \in C^2(I)$ for a maximal interval I where u satisfies the differential equation.

REMARKS.

1. Given real numbers x_0, v_0, u_0 , there exists a maximal interval $I \subseteq \mathbb{R}$ containing x_0 and a unique $C^2(I)$ function u satisfying (3.2) and has $u'(x_0) = v_0, u(x_0) = u_0$. This follows from the Picard-Lindelöf theorem.
2. Moreover, if $u \in C^2(I)$ satisfies (3.2) for some interval I with $u'(x_0) = 0$ and $g(u(x_0)) = 0$, then $u \equiv u(x_0)$ over I . Indeed, note that the constant function $u(x_0)$ is a solution to (3.2) since $g(u(x_0)) = 0$. By uniqueness, we have $u \equiv u(x_0)$.
3. Suppose $u \in C^2(I)$ solves (3.2) on I and $x_0 \in I$ is fixed. Set $G(s) := \int_{u_0}^s g(\sigma) d\sigma$ for real s . Then, if we write $u_0 := u(x_0)$

$$\frac{1}{2} u'(x)^2 + G(u(x)) = \frac{1}{2} u'(x_0)^2 + G(u(x_0)) \quad (3.3)$$

Certainly, by our regularity condition the function u' is non-degenerate and hence

$$u' u'' + u' g(u) \equiv 0, \quad \text{on } I$$

whereby,

$$\int_{x_0}^x (u' u'' + u' g(u)) dx = 0$$

which yields the result.

4. Let $I = (x_0 - h, x_0 + h)$ for $h > 0$ denote an interval about x_0 and suppose $u \in C^2(I)$ satisfies (3.2) on I with $u'(x_0) = 0$. Then u is symmetric about x_0 . That is, for all $0 \leq s < h$ one has $u(x_0 - s) = u(x_0 + s)$. Certainly, define $v(s) := u(x_0 - s)$ and $w(s) := u(x_0 + s)$ for $s \in (-h, h)$. It is clear that $v'(0) = w'(0) = 0$ and we may apply remark 2 to deduce that $v \equiv v(0)$ and $w \equiv w(0)$ for $-h < s < h$. However, since $v(0) = w(0)$ we deduce by uniqueness that $v \equiv w$ there and we are done.

A consequence of this last remark is that if u is C^2 and solves (3.2) on $(a, b) \subseteq \mathbb{R}$ and u' vanishes at two distinct points $x_1, x_2 \in (a, b)$ then u is periodic on (a, b) with period $\omega := 2|x_2 - x_1|$.

3.1 THE PROBLEM ON THE REAL LINE

Now we handle a specific case of the above equation. Nowhere did we assume $(a, b) \subsetneq \mathbb{R}$, and so the earlier remarks certainly still hold. Here, we study the properties of $u \in C^2(\mathbb{R})$, **real-valued**, that satisfy:

$$u'' + g(u) \equiv 0, \quad \text{in } \mathbb{R} \quad (3.4)$$

together with the decay condition

$$\lim_{x \rightarrow \pm\infty} u(x) = 0 \quad (3.5)$$

LEMMA 3.1.1. *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then, if $f'(c) > 0$ for $c \in (a, b)$ then $f(x) > f(c)$ in a neighbourhood of c . Similarly, if $f'(c) < 0$ then $f(x) < f(c)$ in a neighbourhood of c .*

Proof. Suppose without harm that $f'(c) > 0$, replacing f with $-f$ otherwise. Then, take $\varepsilon := f'(c)/2 > 0$, whereby for some $\delta > 0$

$$\left| \frac{f(x) - f(c)}{x - c} - f'(c) \right| < \frac{f'(c)}{2}, \quad \text{whenever } 0 < |x - c| < \delta$$

In other words, for such x :

$$0 < \frac{f'(c)}{2} < \frac{f(x) - f(c)}{x - c}$$

Hence, if $x - c > 0$ then $f(x) - f(c) > 0$. This is guaranteed in $(c, c + \delta)$. The lemma is proven. ■

THEOREM 3.1.2 (Darboux). *Let I be an open interval and $a, b \in I$ and suppose f is continuous on the closure of I . Assume that f is real valued and differentiable on I . If $k \in (f'(a), f'(b))$, without order, then there exists $c \in (a, b)$ so that $f'(c) = k$. That is, derivatives have the intermediate property.*

Proof. Let $g(x) := kx - f(x)$. This function is also differentiable on I . Suppose without loss of generality that $f'(a) < k < f'(b)$, relabeling otherwise. g must achieve a maximum on $[a, b]$, since this is a compact interval and is contained in the closure of I where f is continuous. Note that by hypothesis $g'(a) > 0$ and $g'(b) < 0$. Therefore, the maximum of g is neither achieved at a nor b . In other words, the maximum on $[a, b]$ is a local maximum, where g' vanishes. ■

When convenient, we shall denote the extended real line by $\overline{\mathbb{R}} := [-\infty, \infty]$. We now give a generalization of Rolle's theorem:

THEOREM 3.1.3 (Generalized Rolle's Theorem). *Let $[a, b] \subseteq \overline{\mathbb{R}}$ and $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = f(b) = 0$, then there exists a point $x_0 \in (a, b)$ so that $f'(x_0) = 0$.*

Proof. If f is constant, it must be identical to 0 everywhere and so take any finite point. Otherwise, let $[a, b] \subseteq \mathbb{R}$. Suppose without loss of generality that f achieves positive value at some interior point c of (a, b) . Then, there must exist a point $c_* \leq c$ where $f'(c_*) \geq 0$. If $f(c_*) = 0$ we are done, otherwise it is positive. Similarly, since $f(b) = 0$ and f achieves positive value at $c < b$ then by differentiability we must be able to find a point $c^* > c$ so that $f'(c^*) \leq 0$. If $f'(c^*) = 0$ we are done. Otherwise, we have found two points with $f'(c^*) < 0 < f'(c_*)$, whence the *Darboux Theorem* guarantees suitable x_0 .

■

LEMMA 3.1.4. *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuously differentiable on $\mathbb{R}$, set $L > 0$ and suppose that $\lim_{x \rightarrow \infty} f'(x) = L$, then $\lim_{x \rightarrow \infty} f(x) = \infty$.*

Proof. By l'Hopital's rule we know that

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = L > 0$$

Now let $N > 0$ be given. By the limit above, there exists $M_1 \gg 0$ so large that for all $x \geq M_1$ one has

$$\frac{L}{2} \leq \frac{f(x)}{x}$$

Now let $M_2 \gg 0$ so large that $ML/2 \geq N$. Then, for all $x \geq \max(M_1, M_2)$:

$$f(x) \geq \frac{xL}{2} \geq \frac{M_2L}{2} \geq N$$

which proves the claim.

■

COROLLARY 3.1.5. *Fix $x_0 \in \mathbb{R}$ and let $u \in C^2([x_0, \infty))$. Suppose $u(x) \rightarrow \ell \neq \pm\infty$ and $u''(x) \rightarrow 0$ as $x \rightarrow \infty$. Then $u'(x) \rightarrow 0$ as $x \rightarrow \infty$.*

Equipped with some of these preliminaries we can now begin to characterize non-trivial solutions to (3.4)-(3.5). By non-trivial, we simply mean that $u \not\equiv 0$ for all $x \in \mathbb{R}$, although as we shall see even this trivial solution satisfies (weakly) much of the following as a special case:

THEOREM 3.1.6. *Let $u \in C^2(\mathbb{R})$ be a non-trivial solution to (3.4)-(3.5). Then,*

- (I) $g(0) = 0$.
- (II) *There exists a unique point $x_0 \in \mathbb{R}$ so that $u'(x_0) = 0$. Moreover, if u is positive at some point then u is symmetric about x_0 and u is increasing for $x < x_0$ and decreasing for $x > x_0$. If instead u is negative somewhere then u is decreasing for $x < x_0$ and increasing for $x > x_0$.*
- (III) *Either $u > 0$ or $u < 0$ on $\mathbb{R}$, i.e. it is **single signed**.*
- (IV) *A linear translation of u satisfies the problem. In particular, the solution is not unique.*

Proof. We prove these in several steps:

The first follows from (3.4) and the continuity of u'' . Two applications of Lemma (3.1.4) shows that u'' must vanish at infinity, otherwise contradicting (3.5). Thus, passing to the limit in (3.4) we may use the regularity assumption (continuity of u'') to see that this reduces to

$$\lim_{x \rightarrow \infty} g(u(x)) = g(0) = 0$$

since g is continuous and u vanishes at infinity.

Similarly, since u is non-trivial and vanishes at infinity u is not-periodic. Therefore, by our previous remarks u' can have at most one 0. Our extended Rolle's theorem (Theorem (3.1.3)) we know there exists a (now *unique*) point $x_0 \in \mathbb{R}$ where $u'(x_0) = 0$. By previous remarks, we have that u is symmetric about this point.

To see the monotonicity claims, suppose first that u is positive somewhere on $\mathbb{R}$. Then, since u vanishes at infinity it must be increasing at some $x_* < x_0$. Moreover, by uniqueness

of x_0 we also have $u'(x_*) > 0$. If u were also decreasing for some $x^* < x_0$, then we would have $u'(x^*) < 0$, where by continuity u' would vanish at one more point: contradiction. A verbatim argument holds for $x > x_0$ and for $u < 0$ somewhere in $\mathbb{R}$. This proves (II).

To see (III), we use a similar argument, note that our proof in (II) leads to contradictory results if u switches signs.

(IV) is obvious by simple computation.

■

Our next goal is to give sufficient and necessary conditions for the existence of solutions to the problem in (3.4)-(3.5). Before we do this, we require several lemmata.

LEMMA 3.1.7. *Let u be even and satisfy (3.4)-(3.5), then for each $x \in \mathbb{R}$:*

$$\frac{1}{2}u'(x)^2 + G(u(x)) = 0 \tag{3.6}$$

Certainly, recall that we may write

$$\frac{1}{2}u'(x)^2 + G(u(x)) = \frac{1}{2}u'(x_0)^2 + G(u(x_0))$$

for each $x_0 \in \mathbb{R}$. Now, we let $x_0 \rightarrow \infty$ and the continuity of u' and $G(s)$.

THEOREM 3.1.8 (Schrödinger's Criterion).